

# KARAN CHAWLA DORA

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## EXPERIENCE

### SAE AutoDrive - Toronto Autonomous Vehicle Team (aUToronto)

Sep. 2025 – Present

State Estimation Developer

Toronto, ON

- Contributed in creating a **lane localization** model to improve vehicle positional accuracy under GPS attenuation.
- Leveraged **ROS** for data analysis and testing.

### FIRST Robotics Competition

Jul. 2021 – Jun. 2025

Robotics Team Captain & Software Development Lead

Ottawa, ON

- Founding member and leader of a 90+ high school student team competing at a national level, qualifying for the **World Championship** and **2x Finalists at Provincial Championships**.
- Developed accurate autonomous routines in **C++**, leveraging **computer vision** and **odometry**.
- Enhanced robot performance and precision using feedforward and PID feedback **control systems**.

## RESEARCH & PUBLICATIONS

### Researcher @ Autonomous Space Robotics Lab (ASRL) | ROS, RF, State Estimation

May 2026 – Present

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### Bruce Lab — University of Ottawa | Research Assistant

Aug. 2024

- Reduced transient time by 80% and steady-state error by 40% by implementing PID control systems in C++ to stabilize current and voltage in platinum-nickel electrodeposition for Fuel Cell Electric Vehicles.
- Reduced calibration time by one-third, developing a Python user interface to conduct trials **autonomously**.
- Expanded trial voltage limits by 5x, refining an **Arduino**-based potentiostat electric circuit.

### Canadian Robotics and Artificial Intelligence Ethics Design Lab | Robotics Research

Aug. 2024

- Accelerated project timeline by 50% by characterizing DJI's Robomaster S1 in a simulative environment.
- Achieved 1-second target tracking by implementing computer vision models with Python **machine learning**.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Java, Javascript, Typescript, HTML, CSS

**Robotics & Control:** ROS, PID control, State Estimation, Sensor Fusion

**Tools & Libraries:** NumPy, Matplotlib, pandas, Arduino, Git, Next.js

## AWARDS

**HackTheNorth 2024 2x Award Winner:** Best Use of MongoDB, SymphonicLabs

**2x FRC Provincial Finalists:** FIRST Robotics 2023, 2024

**Harvard-MIT Math Tournament:** Qualified Participant

**FIRST Alumni Entrance Scholarship:** Value of \$15,000

**Dean's List Semi-finalist:** First Robotics Competition

**University of Toronto Excellence Award (UTEA):** For Summer Research Valued at \$7,000

## EDUCATION

### University of Toronto

Toronto, ON

Bachelor of Applied Science — Engineering Science (cGPA: 3.95/4.0)

2025 - 2030

**Research Interests:** Control systems, robotics autonomy, state estimation